

InfEHR: Clinical phenotype resolution through deep geometric learning on electronic health records

Corresponding Author: Professor Girish Nadkarni

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Title: Resolving Clinical Uncertainty through Deep Geometric Learning on Electronic Health Records

General comments:

The study proposed a novel method using deep geometric learning models to render EHR information to Inf-EHR to reduce human pre-specification in computing the likelihood of prediction. The neonatal culture-negative sepsis (CN-S) representing a rare disease and adult post-operative acute kidney injury (PO-AKI) representing a common disease were selected to compare the performance between the proposed model and physician-provided clinical heuristics. The following issues are suggested:

- 1) A baseline characteristics table should be included to clarify the study population. Key demographics and clinical variables for both CN-S and PO-AKI cohorts would provide context and allow for better comparison between the training and validation datasets.
- 2) It is suggested to reveal the number of patients excluded during the data preprocessing process in the training and validation datasets for CN-S and PO-AKI cohorts. individually.
- 3) To help the readers understand how the proposed model performed in the rare and common diseases selected by the study, the positive rate in the CN-S and PO-AKI the training and validation datasets are suggested to be reported, respectively.
- 4) The description and performance of validation dataset for CN-S were not provided (Line 965-969 and Table 1). It is not acceptable to only demonstrate the performance of training data.
- 5) Figure 2a, and the second graph in Figure 5b compared the performance of GNN, ML and Clinical heuristics for CN-S with duplicated information thus Figure 2a should be deleted. Similarly, Figure 3a and the first graph in Figure 5b contained same information and one of them should be deleted.
- 6) The study proposed that the inf-EHR framework naturally captures phenotypic temporal dynamics with a few labeled data compared with the machine learning (ML) based prediction model. It would be useful to compare the model performance and the number of labeled datasets with the ML based approach in Table 1, as the Figure 5b indicated the ML model have been developed.
- 7) Adding a workflow diagram in the methods section would improve readability with a well-defined hierarchy. The method section should be reorganized into distinct subsections.

Minor suggestions:

- 8) It is also suggested to check the consistency of terminology throughout the manuscript. For example, GNN and Inf-HER are interchanged in Table 1, or UCIMC data and MOVER.
- 9) Abbreviations are not well explained when firstly used or in the Table or Figures. For example, MSHS and UCIMC in Table 1 and Line 1256, or ROS in Figure 2.
- 10) Some graphs axis labels, annotation, and footnotes are missing, or have misplaced units. For example, Figure the axis label of 1a, 1c, 2b, 2c, are missing. The axis and unit for C-Reactive protein in Figure 2b are misplaced. The annotation of Figure 1c is incomplete. It is also suggested that the color of Figure 2c to be adjusted to clearly demonstrate the data.

(Remarks on code availability)

The code are reviewed and can be used to duplicate the study in similar structural EHR data.

Reviewer #2

(Remarks to the Author)

This paper presents a novel method for making predictive modelling for EHR time series data.

Strong points:

- * Large scale data set used
- * Two relevant prospective test scenarios with one rare event neonatal culture-negative sepsis and one common event adult post-operative acute kidney injury
- * Establish baseline heuristics to compare the developed approach against
- * Report good performance
- * Code available on Github
- * Good figures

Weak points:

- * Paper very informal in its style. For example, the central concept of likelihoods are mentioned 84 times in the paper, but not given a concise definition.
- * The formatting is problematic especially with regards to equations and algorithms that are presented as a mix of in text in word format and screenshots from latex.
- * Important: Lack of benchmarking against other similar in spirit deep learning based approaches. A list of references are given below.

Conclusion: Paper holds promise but has to be improved in its presentation style and benchmarking.

References:

Life to vec <https://www.nature.com/articles/s43588-023-00573-5>

Gru-d <https://arxiv.org/abs/1606.01865>

Seft <https://arxiv.org/abs/1909.12064>

(Remarks on code availability)

Have not run the code, but I have looked through the repo and it definitely looks well-organised and thorough.

Reviewer #3

(Remarks to the Author)

Dear authors!

Thank you for sharing your in depth investigation on this topic. Here is some feedback from my side:

[A] When you extract the clinical events from the Mount Sinai Data Warehouse, how do you deal with events of a patient which are just documented in clinical narratives (clinical progress notes, semi-structured, non-standardized), typically in EHRs? Can you elaborate on this? Did you use an NLP approach for this initial data extraction progress, in addition to what is already documented in a structured and even standardized way?

[B] For coding your events, did you use international terminology standards like SNOMED CT? Do you have an overview of the distinct set of events that occurred in the uses cases?

[C] I understood that out of a labelled set of 110 cases you inferred a methodology how to label the remaining ones (MAP@1 CNS: 0.79, MAP@1 PO-AKI: 0.71) which can be used for representational learning and GNN tuning. You said in there you took the nearest neighbour to assign the label. Would it be an idea to parametrize the distance in a way (e.g. >0.9), that you are more confident with the assigned label, though having some false negatives then? Though reporting on the quality of this process, do you feel confident that with these numbers (MAP@1 CNS: 0.79, MAP@1 PO-AKI: 0.71) the ongoing investigation is stable in modelling the information content based on the weakly labelled data?

[D] The chosen hyperparameters should be stated at some point in the paper when tuning the GNN for the tasks under investigation. What was the embedding dimensionality?

[E] "We observed that many such automatically generated heuristics are corroborated by findings in literature (see Supplementary Materials for specific examples)." What about the other way around, that the structural heuristics can be inferred out of literature as additional "clues"?

(Remarks on code availability)

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Several major issues have been largely revised according to the previous suggestions. The readability has been improved. However, there are some adjustments required and stated as follows,

- 1) The previous suggestions on the first and third items have not yet been fully addressed in this revision. Supplemental Table 1 provides the number of PO-AKI patients in each subgroup for UCIMC and MSHS. Please also add a row for the total number, percentage, etc. Although the data was provided for PO-AKI, the statistical comparison of the positive rate for PO-AKI was not performed. Limited features and variables were provided for the PO-AKI and CN-S.
- 2) The CN-S cohort has no external validation. This should be mentioned as a limitation in the discussion section.
- 3) The manuscript uses "we" more than one hundred times; revision is recommended to enhance objectivity in academic writing.
- 4) It is not recommended to begin a sentence with Arabic numerals in formal academic writing. Please check and revise, as some sentences in the manuscript have this issue.

(Remarks on code availability)

The codes are readable and can be well understood.

Reviewer #2

(Remarks to the Author)

The authors have shown a strong willingness to revise the paper according to the suggestions from the reviewers. The paper is much improved now. This reviewer for example much appreciates the improvements in the presentation (still recommend working even more on the being more concise) and on the benchmarking against existing methods.

The proposed approach still have many moving parts and appears very engineered (e.g. in the NLP part that appear old school compared to the more generalist approaches prevailing in the LLM age) so the proposed method will not become a standard, but rather served as inspiration for the development going forward.

(Remarks on code availability)

I assessed the code in the first round.

Reviewer #3

(Remarks to the Author)

Dear authors!

Thank you for addressing my comments and issues in detail, which are clarified from my point of view now. From my side, I will suggest acceptance of this version of the manuscript.

(Remarks on code availability)

Open Access This Peer Review File is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made.

In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.

The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>

InfEHR: Resolving Clinical Uncertainty through Deep Geometric Learning on Electronic Health Records

Response to the reviewers

REVIEWER COMMENTS

Reviewer No. 1

1) A baseline characteristics table should be included to clarify the study population. Key demographics and clinical variables for both CN-S and PO-AKI cohorts would provide context and allow for better comparison between the training and validation datasets.

Response: We appreciate the comments from the reviewer and the effort of reviewing. We have included two new tables as shown below covering cohort details for MSHS and UCIMC data for the PO-AKI prediction task. In the tables we include clinical variables such as age, and sex, and ASA rating (a common benchmark used by anesthesiologists to assess risk). We additionally provide correlation coefficients between surgery types and outcomes to provide additional points of comparison between the MSHS and UCIMC cohorts. We also include clinically relevant demographic variables for the Culture-Negative Sepsis cohort organized by training and validation split.

Table S1: Clinical Settings for UCIMC and MSHS cohorts (PO-AKI)

Category	Total Count		Positive Count		% of Total Positives		Correlation		P-value	
	UCIMC	MSHS	UCIMC	MSHS	UCIMC	MSHS	UCIMC	MSHS	UCIMC	MSHS
Urological	255	97	122	21	36.86	2.39	0.348	0.004	6.33e-76	8.02e-01
Cardiovascular	413	1027	91	354	27.49	40.32	0.123	0.193	2.45e-10	5.97e-37
Transplant	1	38	0	34	0.00	3.87	-0.007	0.161	7.04e-01	2.93e-26
Gastrointestinal	192	1293	20	180	6.04	20.50	-0.018	-0.110	3.48e-01	7.82e-13
Pulmonary and Respiratory	115	193	9	39	2.72	4.44	-0.031	-0.002	1.16e-01	8.82e-01
Neurosurgical	255	151	11	17	3.32	1.94	-0.082	-0.044	2.74e-05	3.75e-03
Orthopedic	617	307	35	47	10.57	5.35	-0.115	-0.037	3.01e-09	1.68e-02
General Surgery	783	1150	43	186	12.99	21.18	-0.139	-0.067	7.58e-13	1.22e-05

Note: Correlation values are point-biserial correlations. P-values below 0.05 indicate statistical significance. Correlation interpretation (Evans, 1996): 0.00–0.19 (very weak), 0.20–0.39 (weak), 0.40–0.59 (moderate), 0.60–0.79 (strong), 0.80–1.00 (very strong)

Table S2: Demographic and Clinical Characteristics of UCIMC and MSHS Cohorts (PO-AKI)

Characteristic	UCIMC	MSHS
Sex (Count)		
Male	1,519	2,258
Female	1,070	1,998
Age		
Median (IQR)	59 (44–72)	65 (53–75)
Range	17–90	18–89
ASA Score		
Median (IQR)	3 (3–4)	3 (3–4)
Range	1–5	1–5

Note: IQR = Interquartile Range (Q3–Q1).

Table S3: Distribution of Clinical Variables in MSHS Training and Validation Sets (CN-S)

Variable	Minimum		Maximum		Median		IQR	
	Training	Validation	Training	Validation	Training	Validation	Training	Validation
Birth Weight (oz)	11.46	14.11	180.25	140.74	58.12	58.12	58.21	55.25
Gestational Age (weeks)	22.00	23.00	43.30	41.10	32.00	32.00	11.20	11.00
Age (days)	0.00	0.00	342.00	235.00	2.00	3.00	23.00	24.25
Corrected GA (weeks)	22.00	23.50	83.30	69.10	34.50	35.00	10.10	8.95
GA (days)	154.00	161.00	304.00	288.00	224.00	224.00	79.00	77.00
Apgar (1 min)	0.00	1.00	9.00	9.00	7.00	7.00	4.00	4.00
Apgar (5 min)	0.00	1.00	9.00	9.00	8.00	8.00	3.00	2.00
Apgar (10 min)	0.00	2.00	10.00	9.00	7.00	8.00	1.25	1.00

Note: IQR = Interquartile Range (Q3–Q1). Total cohort size: 3,643 patients. Training: n=2,914 (84 positives). Validation: n=729 (53 positives). Split stratified by Birth Weight.

2) It is suggested to reveal the number of patients excluded during the data preprocessing process in the training and validation datasets for CN-S and PO-AKI cohorts. Individually.

Response: We appreciate this comment by the reviewer. We have made this information clearer in the methods section excerpted below:

“EHR Preprocessing

We apply pre-processing steps described below to the CN-S and PO-AKI training datasets (individually) and apply the results where applicable to the validation datasets. The Mount Sinai Hospital System consists of several individual hospitals with varying database capture and update protocols. As a result, some types of information were systematically unavailable in the data warehouse at the time of retrieval. EHRs with such categorical missingness (e.g., no vitals) were excluded from model training given that this pattern of missingness was likely due to database specific variation (retained cases: n=5213 CN-S, n=4276 PO-AKI MSHS, excluded cases: n=2854 CN-S, n=3764 PO-AKI MSHS) however we use all valid records, including incomplete records, for the density estimations required by the node discovery process (see Method of Node Discovery below).”

3) To help the readers understand how the proposed model performed in the rare and common diseases selected by the study, the positive rate in the CN-S and PO-AKI the training and validation datasets are suggested to be reported, respectively.

Response: We have included this information in the tables produced above in (1).

4) The description and performance of validation dataset for CN-S were not provided (Line 965-969 and Table 1). It is not acceptable to only demonstrate the performance of training data.

Response: To address the reviewer’s critical concerns, we performed an 80/20 Testing / Validation stratified split. On advice from a neonatologist, we stratified the splits based on birth weight as this feature is singular, label agnostic, and provides an important clinical context for predictions. The splits result in different prevalences (.0288 Train, .0728 Test). We provide more information about underlying clinical differences in the table produced above (in 1). Despite the underlying clinical differences and prevalence rates the InfEHR GNN nonetheless yields high rule-in potentials in both cases (16.02 Train, 12.49 Validation) while maintaining comparable Rule-Out potentials (2.88 Train, 2.43 Validation).

While we acknowledge the importance of validation performance, we note that our primary objective with InfEHR aligns with uncertainty resolution frameworks like that presented in Rolf et al. (2022), where the goal is to demonstrate a model's ability to take

uncertain probabilistic inputs and improve their quality through learned revisions. The training performance we show demonstrates InfEHR's capacity for this uncertainty resolution task, while the validation results confirm generalization to unseen cases. We thank the reviewer for pointing out the deficiency in our original submission and hope that we have now appropriately presented InfEHR's performance.

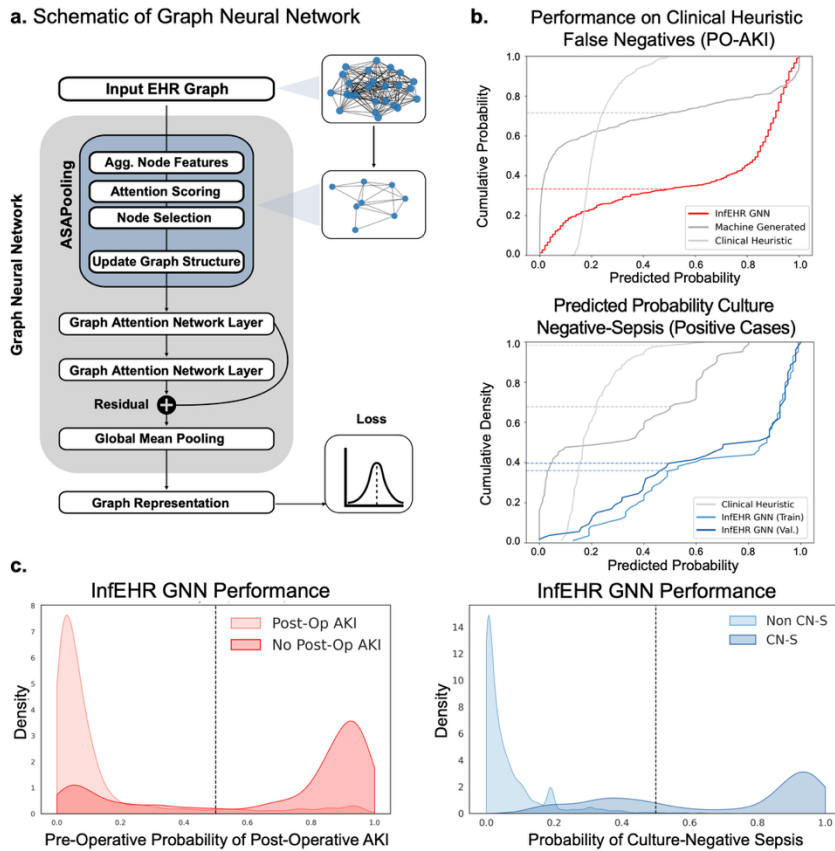
Table 1: InfEHR Performance: CN-S (MSHS)

Metric	Clinical Heuristic	Input Priors (Machine Generated)	InfEHR GNN (Train)	InfEHR GNN (Test)
Prevalence	0.0376*	–	0.0288	0.0728
Threshold	0.500	–	–	–
Sensitivity	0.041	0.524	0.634	0.603
Specificity	0.983	0.990	0.995	0.995
PPV	0.31	0.620	0.830	0.914
NPV	0.85	0.982	0.986	0.969
Rule-In	1.133	5.617	16.018	12.492
Rule-Out	1.007	1.984	2.880	2.435

* Whole dataset(n=5213 cases), n=44 CN-S positives identified by keyword, n=3643 cases manually reviewed to identify a total of n=137 positive cases (Prevalence = 0.028). We use only manually confirmed cases for training, testing, and reporting of supervised metrics (Prevalence=0.0376).

5) Figure 2a, and the second graph in Figure 5b compared the performance of GNN, ML and Clinical heuristics for CN-S with duplicated information thus Figure 2a should be deleted. Similarly, Figure 3a and the first graph in Figure 5b contained same information and one of them should be deleted.

Response: We acknowledge the significant overlap of these figures. To address this, we provide new figures with more distinctive information. In 5b (1) we have revised the figure to show the training and testing performance over positive cases vs. 2a, which shows the heuristic performance on all positive and negative cases. In 5b(2) we have revised the figure to show only the performance of InfEHR on false negative cases according to the clinical heuristic.



6) The study proposed that the inf-EHR framework naturally captures phenotypic temporal dynamics with a few labeled data compared with the machine learning (ML) based prediction model. It would be useful to compare the model performance and the number of labeled datasets with the ML based approach in Table 1, as the Figure 5b indicated the ML model have been developed.

Response: We appreciate this comment by the reviewer. To address this suggestion, we highlight the performance differences between the GNN and the machine generated heuristics in the updated figures (5b panel 1 and 2 see above) as well as in Table 1 above and Table 2 below. We also include benchmarks against SeFT and GRU-D which provides information about the performance of the InfEHR GNN (a geometric model) against non-geometric deep learning models.

Table 2: InfEHR Performance: PO-AKI (MSHS)

Metric	Clinical Heuristic	Machine Generated Priors	InfEHR GNN
Prevalence	0.228*	0.228	0.228
Threshold	0.500	–	0.500
Sensitivity	0.197	0.528	0.706
Specificity	0.978	0.930	0.930
PPV	0.695	0.690	0.724
NPV	0.824	0.869	0.924
Rule-In	1.322	2.110	3.613
Rule-Out	1.161	1.630	2.481

* Whole dataset n=8031, n=1892 positives

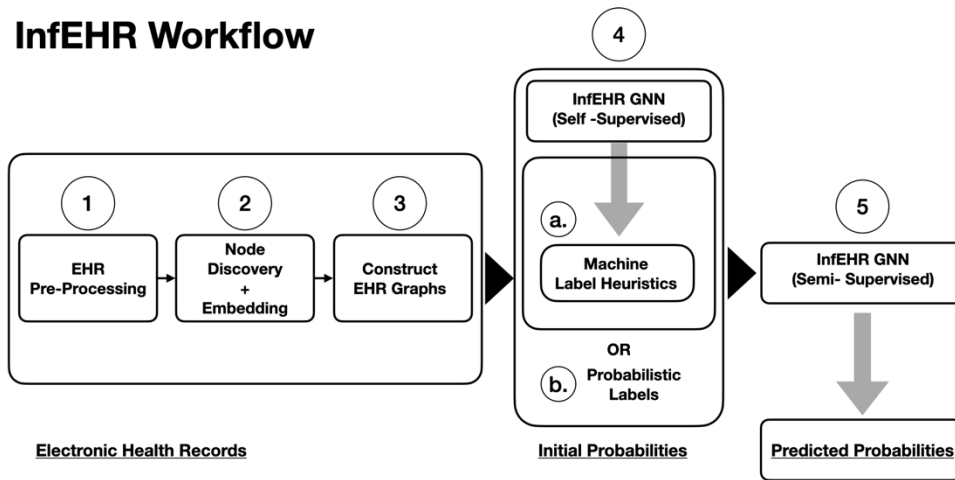
PPV: Positive Predictive Value

NPV: Negative Predictive Value

7) Adding a workflow diagram in the methods section would improve readability with a well-defined hierarchy. The method section should be reorganized into distinct subsections.

Response: We have implemented this suggestion and thank the reviewer for pointing out where we can improve readability. Please find the workflow diagram below (the methods section has been updated to follow this workflow with matching subsection names).

InfEHR Workflow



Minor suggestions:

8) It is also suggested to check the consistency of terminology throughout the manuscript. For example, GNN and Inf-HER are interchanged in Table 1, or UCIMC data and MOVER.

Response: We appreciate the reviewer's astute observation and have revised the manuscript to refer to data obtained from MOVER (a named dataset) as UCIMC (the location of its origin) to be consistent with Mount Sinai obtained data as MSHS in all instances. We have also followed the reviewer's advice and added clarification to all InfEHR references.

9) Abbreviations are not well explained when firstly used or in the Table or Figures. For example, MSHS and UCIMC in Table 1 and Line 1256, or ROS in Figure 2.

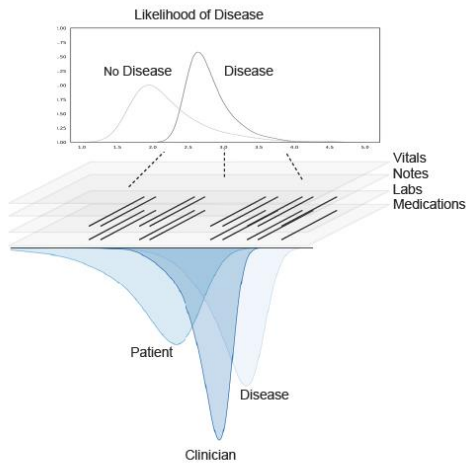
Response: We again thank the reviewer for this observation and have made sure to introduce all abbreviated terms prior to using the abbreviation. This will greatly enhance readability.

10) Some graphs axis labels, annotation, and footnotes are missing, or have misplaced units. For example, Figure the axis label of 1a, 1c, 2b, 2c, are missing. The axis and unit for C-Reactive protein in Figure 2b are misplaced. The annotation of Figure 1c is incomplete. It is also suggested that the color of Figure 2c to be adjusted to clearly demonstrate the data.

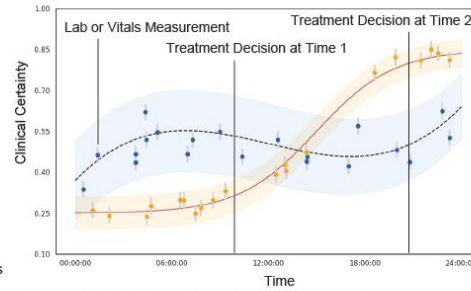
Response: We appreciate the reviewer for pointing out these important details. We have added axis labels in 1c and added annotations, corrected the units in 2b from

mg/dL to mg/L. The units for the counts indicated by the y-axis are included in the figure legend. We have adjusted the colors in 2c to be more distinctive. We omit units in 1a due to the small size of the axis and its intention as a pictorial. In 2c we display the densities of antibiotic course durations but to enhance the visual display of the information include no y-axis. We indicate the number of elapsed hours in the x-axis (see below on document page 9).

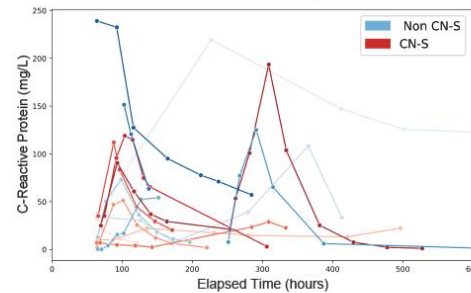
a. Electronic Health Records As Evidence



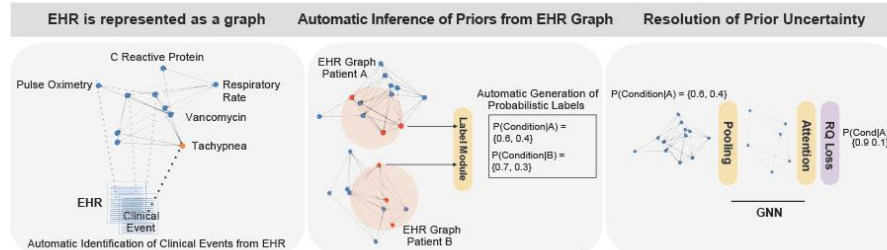
b. Evolution of Clinical Certainty Over Time



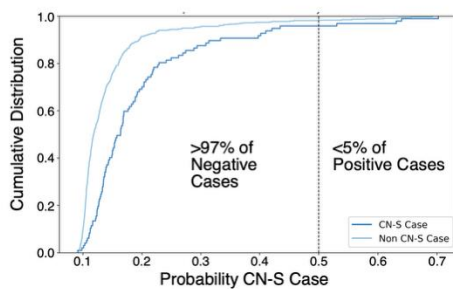
c. Neonatal C-Reactive Protein Dynamics



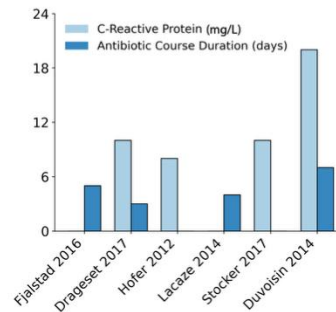
d.



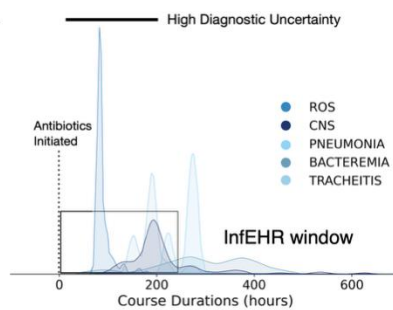
a. Performance of Clinical Heuristic



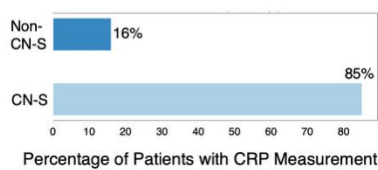
b. Published Guidelines for Diagnosis Culture-Negative Sepsis



c.



d. C-Reactive Protein Measurement in NICU Patients



Reviewer No. 2 (Remarks to the Author).

Strong points:

- * Large scale data set used
- * Two relevant prospective test scenarios with one rare event neonatal culture-negative sepsis and one common event adult post-operative acute kidney injury
- * Establish baseline heuristics to compare the developed approach against
- * Report good performance
- * Code available on Github
- * Good figures

Response: We appreciate the reviewer's positive comments and the effort in reviewing the manuscript. Please find below a point-by-point response to the remaining comments.

Weak points:

1. Paper very informal in its style. For example, the central concept of likelihoods are mentioned 84 times in the paper, but not given a concise definition.

Response: We appreciate this point from the reviewer. The reviewer is right to suggest that improving the formality would strengthen the manuscript. In response, we have removed the term likelihood where it may have been used less formally (or even somewhat inappropriately). We wished to convey to the reader that probabilities produced by models are of a different caliber than those obtained from e.g. a probability density function. Namely that probabilities obtained from models are inescapably estimates (but ideally very good estimates). We have replaced our previous usage where needed with the term “predicted probability” and provided a few sentences to concisely define this as below:

“We train the InfEHR GNN under a specialized loss function similar to [72], which implicitly models the likelihood $P(EHR|Condition)$ following a generative modeling framework. This approach ultimately yields predicted probabilities over conditions given EHR data in the form of a posterior distribution. This mirrors in mathematical terms the colloquial use of the phrase 'likelihood of disease'. Formally, while the InfEHR GNN outputs what is structurally a discriminative posterior $P(Condition|EHR)$, it inherits generative modeling characteristics through our training procedure, which minimizes the Kullback-Leibler divergence between the model output and a theoretically motivated generative posterior $P_{generative}(Condition|EHR) \propto P(EHR|Condition) \cdot P(Condition)$. This infuses our predicted probabilities with the uncertainty-resolving benefits of the generative approaches without requiring explicit generative modeling.”

2. The formatting is problematic especially with regards to equations and algorithms that are presented as a mix of in text in word format and screenshots from latex.

Response. We agree and apologize for the mixture of formatting. We have converted all algorithms and equations to MathML and rendered into MSWord. Please see below for an example of a MSWord rendered equation (screenshot provided, equation inserted as text):

$$\text{Time2Vec}\left(t\right)=\begin{cases}w_k\cdot t+b_k & \text{if }k=0\\ \sin(w_k\cdot t+b_k) & \text{if }k>0\end{cases}$$

where:

- t is the time component (like timestamp, hour of day, etc.).
- w_k and b_k are learnable parameters of the model.
- k refers to the position in the Time2Vec vector.

3. Important: Lack of benchmarking against other similar in spirit deep learning-based approaches. A list of references are given below.

Response: We appreciate this critical insight, and we have performed benchmarking using the UCIMC dataset.

A. GRU-D

This is a very good suggestion as it approaches the problem from a sequential and distinctly non-geometric perspective. We have taken the reviewer's advice and implemented this model and performed benchmarking experiments (see Table 2).

B. SeFT

We also appreciate this suggestion as the architecture has some correspondence to the geometric perspective (since all graphs can be converted to sets). We have performed the requested benchmarking using this model as well (see Table 2).

C. Life2Vec

While we acknowledge the significance of this work some of its more remarkable aspects include the integration of health information with sociological and occupational information (such as education, occupation, and income level) which is not readily available in any of our experiment settings (in particular CN-S where neonates have not yet enrolled in higher education etc.). There is also an incompatibility with the health information considered which appears to be primarily diagnosis codes. We do not use diagnosis codes as they are often applied post-hoc or in a time frame that is

inconsistent with the potential clinical applications we consider. Integrating information from outside the clinical setting is a vital and intriguing suggestion which we hope to pursue in future work but for the reasons outlined above we did not perform benchmarking with this model.

SeFT:

[97] Horn, M., Moor, M., Bock, C., Rieck, B., & Borgwardt, K. (2020). Set Functions for Time Series. Proceedings of the 37th International Conference on Machine Learning, in Proceedings of Machine Learning Research 119:4353-4363.

GRU-D:

[98] Che, Z., Purushotham, S., Cho, K., Sontag, D., & Liu, Y. (2018). Recurrent neural networks for multivariate time series with missing values. Scientific Reports, 8(1), 6085.

Life2Vec (not included):

Savcicens, G., Eliassi-Rad, T., Hansen, L.K., Mortensen, L.H., Lilleholt, L., Rogers, A., Zettler, I., & Lehmann, S. (2024). Using sequences of life-events to predict human lives. Nature Computational Science, 4(1), 43-56.

Please see the table attached below at (4) for results.

4. Conclusion: Paper holds promise but has to be improved in its presentation style and benchmarking.

Response: We hope that we have adequately addressed the reviewer's concerns by providing formal definitions and appreciate the insightful suggestions particularly with benchmarking (included below):

Table 3: PO-AKI Prediction Performance (UCIMC)

Model	Default (0.5)	Optimal Rule-In	Optimal Rule-Out
Clinical Heuristic			
Threshold	0.500	–	–
Sensitivity	0.131	–	–
Specificity	0.994	–	–
PPV	0.717	–	–
NPV	0.904	–	–
Rule-In	1.330	–	–
Rule-Out	1.124	–	–
InfEHR prior			
Threshold	0.500	–	–
Sensitivity	0.540	–	–
Specificity	0.982	–	–
PPV	0.783	–	–
NPV	0.947	–	–
Rule-In	4.139	–	–
Rule-Out	1.898	–	–
GRU-D			
Threshold	0.500	0.253	0.061
Sensitivity	0.605	0.739	0.854
Specificity	0.973	0.964	0.888
PPV	0.728	0.715	0.479
NPV	0.953	0.968	0.981
Rule-In	4.745	6.706	4.354
Rule-Out	2.269	3.172	3.964
SeFT			
Threshold	0.500	0.202	0.081
Sensitivity	0.510	0.759	0.839
Specificity	0.980	0.967	0.909
PPV	0.751	0.733	0.525
NPV	0.943	0.971	0.979
Rule-In	3.415	7.671	4.861
Rule-Out	1.778	3.447	3.935
InfEHR GNN			
Threshold	0.500	0.384	0.195
Sensitivity	0.655	0.730	0.829
Specificity	0.979	0.975	0.896
PPV	0.789	0.780	0.492
NPV	0.959	0.968	0.978
Rule-In	7.105	8.322	4.230
Rule-Out	2.700	3.217	3.566

PPV: Positive Predictive Value

NPV: Negative Predictive Value

Reviewer #3 (Remarks to the Author)

[A] When you extract the clinical events from the Mount Sinai Data Warehouse, how do you deal with events of a patient which are just documented in clinical narratives (clinical progress notes, semi-structured, non-standardized), typically in EHRs? Can you elaborate on this? Did you use an NLP approach for this initial data extraction progress, in addition to what is already documented in a structured and even standardized way?

Response: The reviewer makes an insightful point that not all pertinent information is available in structured fields. We used an NLP based approach to surface this information when available. Specifically, we used QuickUMLS to extract UMLS matching terms from the clinical notes. Next, we used TF-IDF followed by NMF to identify terms that were informative for a given patient (specifically, we identified terms that were strongly connected to a topic identified by NMF). Such selected terms ultimately were embedded and used as nodes in the patient level graphs where they applied. We assign the time stamp of the term to be the date of the note. We explain further in methods below:

“We extract terms from clinical notes using QuickUMLS (see pre-processing of non-numerical data above). We weight the collection of the extracted terms using term frequency inverse document frequency (TF-IDF) , then apply non-negative matrix factorization (NMF) with automatic determination of latent topic number (minimum components in the NMF H matrix such that the cophenetic correlation coefficient ≥ 0.90). We analyze the resulting low rank term-weight matrix to identify and retain terms strongly associated with any latent topic ($\geq$ top 10% of distributed topic weights). This procedure simultaneously selects terms based on frequency and informational content to create a data driven vocabulary of clinically meaningful terms from notes for use as graph nodes.”

While this is an imperfect approach and does not explicitly handle the temporality and semantics of an extracted UMLS term (e.g., is it a past reference or a family history reference) we believe that the graph structure allows the term to be regarded as interacting with other clinical information such as from structured data which carries the needed phenotypic information for the likelihood assignment task. We hope to incorporate more NLP approaches and improve this aspect in future work and have mentioned this important area of improvement in the revised manuscript:

“Future work will also include workflow and pre-processing optimizations. Identifying temporal and other semantic information from the terms extracted from clinical notes can add important contextual information to these terms. Similarly, aligning such terms to graph-based ontologies such as SNOMED-CT presents opportunities to transfer information within the ontology but outside of the EHR directly to patient graphs. These pre-processing efforts taken together with improving overall code performance will further efforts to productionize InfEHR.”

[B] For coding your events, did you use international terminology standards like SNOMED CT? Do you have an overview of the distinct set of events that occurred in the uses cases?

Response: We appreciate this point from the reviewer. While SNOMED CT is indeed a valuable source of ontology information we did not ultimately use it for the following reasons:

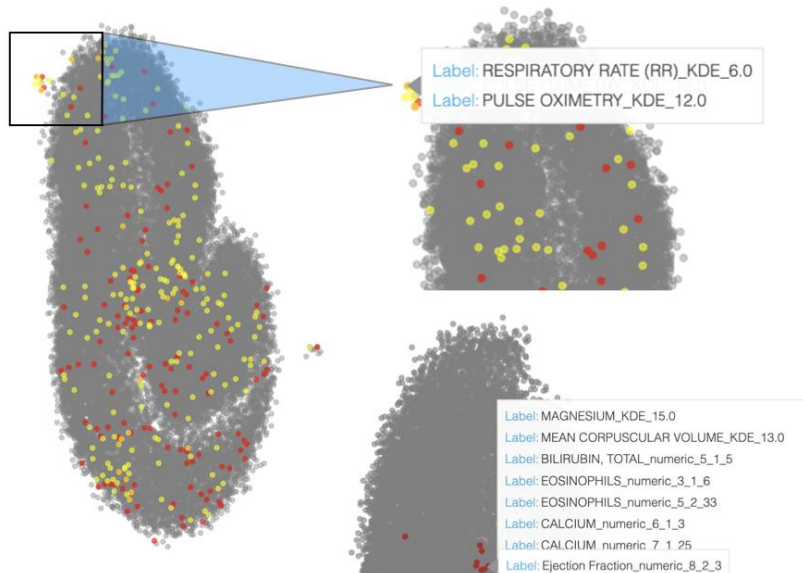
We derive the set of events from the data distribution of the EHRs themselves. For numeric and continuous values this involved applying a kernel density estimation algorithm to identify stable local densities and mapping raw EHR values to this learned space. While we could conceivably map the resulting densities to an ontology structure this may come with a loss of resolution, e.g., systolic blood pressure could be represented through SNOMED CT post-coordination as “271649006 |Systolic blood pressure (observable entity)|” plus a qualifier e.g. “362981000 |Qualifier value (attribute)| = 75540009 |High (qualifier value)|” but this requires that we relate systolic blood pressure values into pre-defined ranges for high, low, etc. We could also relate the raw EHR value into a clinical interpretation using pre-coordination e.g. by mapping to 56218007 | Systolic hypertension (disorder) however here we would also need to apply a priori definitions to the EHRs(e.g. the clinical definition of hypertension). InfEHR provides a framework for working without a priori assumptions or definitions. This allows for clinical events to be extracted from EHRs according to their underlying generative processes which in turn is reflected in the semantics of their embeddings (see Figure 7 attached below for reference).

The reviewer’s suggestion is applicable to the terms we identified from unstructured data (i.e., clinical progress notes). However, the MSHS data warehouse is challenged to map clinical notes to standardized ontologies due to the complexity of the task and computational burden. Clinical notes retrieved from the data warehouse were therefore unmapped and in their native form. We were unable to find a suitable and validated open-source SNOMED-CT mapper for processing these raw notes. To standardize the notes, we used QuickUMLS which is open-source and has been peer-reviewed. The UMLS system used by QuickUMLS does incorporate SNOMED-CT concepts (each SNOMED CT concept receives a UMLS Concept Unique Identifier (CUI)) so while we did not make explicit use of the ontology, we received an implicit benefit. Still, direct use of the SNOMED-CT ontology is an intriguing future research direction as ancestor and descendant concepts can be strategically used to enhance the information content of patient graphs. We thank the reviewer for this suggestion and have mentioned it in future work (see the previous response).

With respect to the distinct set of events in our use cases:

We provide visualizations of the entire clinical manifold consisting of all possible events from which individual patient graphs are composed as a subset in Figure 7 (see detail attached below) Within this manifold, yellow and red highlighted nodes represent event subsets for two distinct patients and events common to both are colored orange. These events are determined from the data distribution of the EHRs as mentioned above without prior specification. As part of the GNN operations, individual graph nodes (representing these events) are coalesced into higher order and compound aggregates.

This results in a new derived graph with nodes that represent something closer to a clinical concept vs. a discrete event. We hypothesize that maintaining high resolution of individual events (e.g., by learning from the EHR data distribution) benefits this operation.



[C] I understood that out of a labelled set of 110 cases you inferred a methodology how to label the remaining ones (MAP@1 CNS: 0.79, MAP@1 PO-AKI: 0.71) which can be used for representational learning and GNN tuning. You said in there you took the nearest neighbour to assign the label. Would it be an idea to parametrize the distance in a way (e.g. >0.9), that you are more confident with the assigned label, though having some false negatives then? Though reporting on the quality of this process, do you feel confident that with these numbers (MAP@1 CNS: 0.79, MAP@1 PO-AKI: 0.71) the ongoing investigation is stable in modelling the information content based on the weakly labelled data?

Response: The reviewer raises an excellent point about implementing distance-based confidence thresholding for our spatial label propagation process. This suggestion touches on a fundamental trade-off in weakly supervised learning between label quality and coverage (patterns that can be captured) that has been systematically studied in the data programming literature. We generate heuristics which either apply or do not apply to an individual case. In the latter scenario, a particular heuristic does not

contribute any information about a particular case, thus not contributing to what Ratner et al. term “label density”, specifically “the mean number of non-abstention labels per data point”.

The proposed approach of using a distance parameter (e.g., >0.9) to retain only high-confidence spatial labels would likely improve the quality and accuracy of our individual extracted subgraph patterns but might come at the expense of the overall label quality as we explain below.

If we implemented confidence thresholding, e.g., only accepting spatial labels where distance <0.1 , we would likely dramatically reduce the number of cases receiving any spatial label, which would cascade to fewer cases available for pattern extraction, fewer extractable heuristics, and ultimately lower label density when these heuristics are applied dataset wide. To better understand the impact of this we turn to the empirical guidelines outlined in Ratner et al concerning the impact of label density on label quality.

They examine the performance of a probabilistic labeling system under high, medium, and low label density regimes. As a baseline, labels can be computed using majority vote of all applicable labeling heuristics. However, majority vote treats all heuristics as equally accurate when they are not. By learning correlation and other dependency structures, the individual accuracy of a labeling heuristic can be estimated which allows for preferential weighting of more accurate heuristics in computing a final label. This leads to an accuracy advantage over majority voting.

In low-density regimes, sparse overlapping predictions prevent meaningful correlation and dependency structures from being learned and therefore provide little benefit over majority vote. Ratner establishes in Proposition 1 that the expected advantage over majority vote has an upper bound that “falls quadratically with label density”. In Proposition 2 Ratner shows that at high label density and average heuristic accuracy $> 50\%$, majority vote converges exponentially to optimal (full knowledge of the ground truth). However, we are unlikely to meet these label densities through simple heuristics given the complex nature of the clinical task and noisiness of EHRs (see the low performance of clinical heuristics as tangible support to this point).

Our current MAP@1 scores (0.79 CN-S, 0.71 PO-AKI) enable generation of sufficient diverse heuristics from our spatial labeling approach to operate in what Ratner et al. term the ‘medium density regime’ (empirically $d \approx 1-3$ labeling heuristics per case) which they show to be suitable for denoising imperfect heuristics and achieving an advantage over a majority vote. Our approach follows Ratner et al.’s core design principle that weak supervision frameworks should be designed to handle imperfect labeling functions

(and correspondingly imperfect labels), and we show through multiple experiments (see Table 1, Table 2, Table 3) that the machine generated heuristics outperform clinical heuristics and that the GNN can profitably use imperfect initial probabilistic labels to achieve high rule-in and rule-out potentials. We acknowledge that improving the MAP@1 scores would be a valuable future research direction and appreciate the insight the reviewer brings to this issue.

For more detailed information we refer the reviewer to the following citations [100, 101 added in manuscript].

Ratner, A., Bach, S.H., Ehrenberg, H., Fries, J., Wu, S., Ré, C. Snorkel: rapid training data creation with weak supervision. The VLDB Journal 29, 709–730 (2020)

Ratner, A., De Sa, C., Wu, S., Selsam, D., Ré, C. Data programming: Creating large training sets, quickly. In: Neural Information Processing Systems (NIPS) (2016)

[D]The chosen hyperparameters should be stated at some point in the paper when tuning the GNN for the tasks under investigation. What was the embedding dimensionality?

Response: We have updated the manuscript to include this information following the definition of the GNN architecture:

“For all experiments we use input node feature dimensions $d=160$ (from $d=32$ for time embedding + $d=64$ node semantic type embedding + $d=64$ node value embedding). We use a node pooling ratio (ρ) of 0.8, and use hidden dimensions $H(1)=256$, and $H(2)=128$. We train on a self-supervised learning (SSL) objective (explained below) to produce a $d=128$ dimensional representation (dimensions chosen from experience) and a $d=2$ dimensional output for semi-supervised training (distribution over binary classes).”

[E] "We observed that many such automatically generated heuristics are corroborated by findings in literature (see Supplementary Materials for specific examples)." What about the other way around, that the structural heuristics can be inferred out of literature as additional "clues"?

Response: This is a fantastic suggestion and an excellent idea for future applications. We think that while this would be a very lucrative approach there are substantial technical difficulties involved in accurately processing and integrating literature into a format suitable for heuristic generation or application and therefore, we are unable to feasibly accomplish this within an acceptable timeline for revisions.

REVIEWER COMMENTS

1. The previous suggestions on the first and third items have not yet been fully addressed in this revision. Supplemental Table 1 provides the number of PO-AKI patients in each subgroup for UCIMC and MSHS. Please also add a row for the total number, percentage, etc. Although the data was provided for PO-AKI, the statistical comparison of the positive rate for PO-AKI was not performed. Limited features and variables were provided for the PO-AKI and CN-S.

Response. We thank the reviewer for identifying these areas for improvement and agree that providing the total number, percentage, and statistical analysis of positivity rate between MSHS and UCIMC data would add beneficial context. We have accordingly revised the table to make the analysis datasets more directly comparable (table reports values for included subjects only) and have supplied the requested rows:

Category	Total Count		Positive Count		% of Total Positives		% of Total Negatives		Correlation		P-value	
	UCIMC	MSHS	UCIMC	MSHS	UCIMC	MSHS	UCIMC	MSHS	UCIMC	MSHS	UCIMC	MSHS
Urological	210	97	80	21	30.65	2.39	6.00	2.24	0.348	0.004	6.33e-76	8.02e-01
Cardiovascular	390	1027	71	354	27.20	40.27	14.73	19.81	0.123	0.193	2.45e-10	5.97e-37
Transplant	1	58	0	35	0.00	3.98	0.05	0.68	-0.007	0.161	7.04e-01	2.93e-26
Gastrointestinal	175	1293	16	180	6.13	20.48	7.34	32.76	-0.018	-0.110	3.48e-01	7.82e-13
Pulmonary and Respiratory	102	193	7	39	2.68	4.44	4.39	4.53	-0.031	-0.002	1.16e-01	8.82e-01
Neurosurgical	230	151	9	17	3.45	1.93	10.20	3.94	-0.082	-0.044	2.74e-05	3.75e-03
Orthopedic	580	307	28	47	10.73	5.35	25.48	7.65	-0.115	-0.037	3.01e-09	1.68e-02
General Surgery	739	1150	50	186	19.16	21.16	31.81	28.38	-0.139	-0.067	7.58e-13	1.22e-05
Total	2427	4276	261	879	100.00	100.00	100.00	100.00	—	—	—	—

Notes: Correlation values are point-biserial correlations. P-values below 0.05 indicate statistical significance.

Correlation interpretation (Evans, 1996): 0.00-0.19 (very weak), 0.20-0.39 (weak), 0.40-0.59 (moderate), 0.60-0.79 (strong), 0.80-1.00 (very strong)

Overall PO-AKI rate: UCIMC 10.75% vs. MSHS 20.56% ($p < 0.0001$, Chi-square=105.40, Odds Ratio=0.47, 95% CI for difference: -11.67% to -7.93%)

2. The CN-S cohort has no external validation. This should be mentioned as a limitation in the discussion section.

Response. The reviewer correctly points out this limitation. We have included the following sentence in the discussion section (in italics) to make this clear:

This observation suggests value in considering temporal dynamics for separating physiologic from pathophysiologic responses [82]. However, identifying such dynamics, which are subject to individual variation within a conserved pattern, at scale exceeds the capacity of the heuristic and is challenging to traditional temporal models [83]. But InfEHR automatically captures informative temporal dynamics without algorithmic or human pre-specification and can also consider interrelating variations between CRP and other biomarkers such as in CBC results which, by themselves, while reportedly low in sensitivity, may provide more information when integrated with other lab results. *The reported results, while promising, have not been externally validated owing to insufficient external data including with confirmed CN-S status.*

3. The manuscript uses "we" more than one hundred times; revision is recommended to enhance objectivity in academic writing.

Response. The reviewer has made good suggestions to improve the overall conformity to academic writing standards in this manuscript. Accordingly, the use of "we" has been reduced by about 30% to achieve a more objective style however in some cases the word "we" is retained to favor the use of the active voice following the style guidelines.

4. It is not recommended to begin a sentence with Arabic numerals in formal academic writing. Please check and revise, as some sentences in the manuscript have this issue.

Response. Following the reviewer's advice the manuscript has been revised for grammatical and clerical errors and to our knowledge there are no cases of sentences beginning with Arabic numerals.

Reviewer No. 2 (Remarks to the Author):

1. The authors have shown a strong willingness to revise the paper according to the suggestions from the reviewers. The paper is much improved now. This reviewer for example much appreciates the improvements in the presentation (still recommend working even more on the being more concise) and on the benchmarking against existing methods.

Response. We thank the reviewer in turn for the thoughtful suggestions which have greatly improved this manuscript. We have shortened the abstract to ~200 words and have made grammatical edits for concision while keeping a level of detail that would satisfy a broad spectrum of readers.

2. The proposed approach still have many moving parts and appears very engineered (e.g. in the NLP part that appear old school compared to the more generalist approaches prevailing in the LLM age) so the proposed method will not become a standard, but rather served as inspiration for the development going forward.

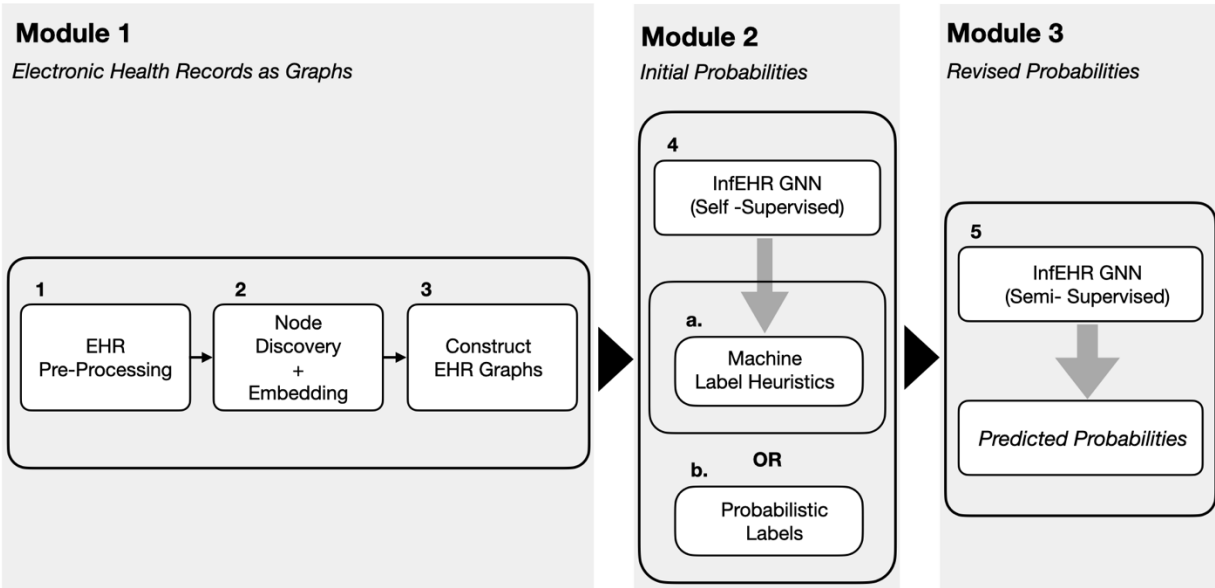
Response. We acknowledge this important point about generalist approaches in the following sentences in our discussion section:

“Future work will include workflow and preprocessing optimizations. Identifying temporal and other semantic information from terms extracted from clinical notes can add important context to these terms. Similarly, aligning such terms to graph-based ontologies such as SNOMED-CT presents opportunities to transfer information from within the ontology but outside the EHR directly to patient graphs. *Large Language Models could potentially be used in preprocessing steps for clinical notes or as embedders for entire clinical notes, which could streamline the existing workflow. The integration of multi-modal information from real-time monitoring or genetics can also be facilitated through generalized models with outputs assimilated by InfEHR. Collectively, these improvements will support bringing the framework into production.*”

Echoing the suggestions in the reviewer’s first comment we have also revised the workflow diagram (See below) to better communicate to the reader that while there are multiple underlying components in the InfEHR framework they can be used efficiently to

render probabilistic inferences. However, we take this suggestion of streamlining as a priority for future work.

InfEHR Workflow



Reviewer No. 3 (Remarks to the Author):

Dear authors!

Thank you for addressing my comments and issues in detail, which are clarified from my point of view now. From my side, I will suggest acceptance of this version of the manuscript.

Response. We really appreciate the reviewer's insightful and stimulating questions which provided opportunities for further examining the manuscript.